



Started on	Tuesday, 4 November 2025, 7:10 PM
State	Finished
Completed on	Tuesday, 4 November 2025, 7:13 PM
Time taken	2 mins 17 secs
Marks	1.00/1.00
Grade	4.00 out of 4.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findDuplicate(int arr[], int n) {
4      int slow = arr[0];
5      int fast = arr[0];
6
7      // Step 1: Detect cycle
8      do {
9          slow = arr[slow];
10         fast = arr[arr[fast]];
11     } while (slow != fast);
12
13     // Step 2: Find entry point (duplicate)
14     fast = arr[0];
15     while (slow != fast) {
16         slow = arr[slow];
17         fast = arr[fast];
18     }
19
20     return slow;
21 }
22
23 int main() {
24     int n;
25     scanf("%d", &n);
26     int arr[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     int duplicate = findDuplicate(arr, n);
32     printf("%d\n", duplicate);
33
34     return 0;
35 }
36

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓

	Input	Expected	Got	
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Tuesday, 4 November 2025, 7:13 PM
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Time taken	19 secs
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Grade	4.00 out of 4.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

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```

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3  int findDuplicate(int arr[], int n) {
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13     // Step 2: Find entry point (duplicate)
14     fast = arr[0];
15     while (slow != fast) {
16         slow = arr[slow];
17         fast = arr[fast];
18     }
19
20     return slow;
21 }
22
23 int main() {
24     int n;
25     scanf("%d", &n);
26     int arr[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     int duplicate = findDuplicate(arr, n);
32     printf("%d\n", duplicate);
33
34     return 0;
35 }
36

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓

	Input	Expected	Got	
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Started on** Tuesday, 4 November 2025, 7:13 PM**State** Finished**Completed on** Tuesday, 4 November 2025, 7:15 PM**Time taken** 1 min 9 secs**Marks** 1.00/1.00**Grade** 30.00 out of 30.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 void findIntersection(int arr1[], int n1, int arr2[], int n2) {
4     int i = 0, j = 0;
5     int found = 0; // To handle no intersection case
6
7     while (i < n1 && j < n2) {
8         if (arr1[i] == arr2[j]) {
9             printf("%d ", arr1[i]);
10            found = 1;
11            i++;
12            j++;
13        } else if (arr1[i] < arr2[j]) {
14            i++;
15        } else {
16            j++;
17        }
18    }
19
20    if (!found)
21        printf("No Common Elements");
22

```



```

23     printf("\n");
24 }
25
26 int main() {
27     int T;
28     scanf("%d", &T);
29
30     while (T--) {
31         int n1, n2;
32         scanf("%d", &n1);
33         int arr1[n1];
34         for (int i = 0; i < n1; i++)
35             scanf("%d", &arr1[i]);
36
37         scanf("%d", &n2);
38         int arr2[n2];
39         for (int i = 0; i < n2; i++)
40             scanf("%d", &arr2[i]);
41
42         findIntersection(arr1, n1, arr2, n2);
43     }
44
45     return 0;
46 }
47
48

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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SAHANA A 2024-CSE ▾

S2

Started on	Tuesday, 4 November 2025, 7:15 PM
State	Finished
Completed on	Tuesday, 4 November 2025, 7:15 PM
Time taken	17 secs
Marks	1.00/1.00
Grade	30.00 out of 30.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 void findIntersection(int arr1[], int n1, int arr2[], int n2) {
4     int i = 0, j = 0;
5     int found = 0; // To handle no intersection case
6
7     while (i < n1 && j < n2) {
8         if (arr1[i] == arr2[j]) {
9             printf("%d ", arr1[i]);
10            found = 1;
11            i++;
12            j++;
13        } else if (arr1[i] < arr2[j]) {
14            i++;
15        } else {
16            j++;
17        }
18    }
19
20    if (!found)
21        printf("No Common Elements");
22

```

```

23     printf("\n");
24 }
25
26 int main() {
27     int T;
28     scanf("%d", &T);
29
30     while (T--) {
31         int n1, n2;
32         scanf("%d", &n1);
33         int arr1[n1];
34         for (int i = 0; i < n1; i++)
35             scanf("%d", &arr1[i]);
36
37         scanf("%d", &n2);
38         int arr2[n2];
39         for (int i = 0; i < n2; i++)
40             scanf("%d", &arr2[i]);
41
42         findIntersection(arr1, n1, arr2, n2);
43     }
44
45     return 0;
46 }
47

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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SAHANA A 2024-CSE ▾

S2

Started on Tuesday, 4 November 2025, 7:16 PM

State Finished

Completed on Tuesday, 4 November 2025, 7:16 PM

Time taken 19 secs

Marks 1.00/1.00

Grade **4.00** out of 4.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int arr[n];
6      for (int i = 0; i < n; i++)
7          scanf("%d", &arr[i]);
8      int k;
9      scanf("%d", &k);
10     int i = 0, j = 1;
11     int found = 0;
12     while (i < n && j < n) {
13         if (i == j) {
14             j++;
15             continue;
16         }
17         int diff = arr[j] - arr[i];
18         if (diff == k) {
19             found = 1;
20             break;
21         } else if (diff < k) {
22             j++;
23         } else {
24             i++;
25         }
26     }
27     printf("%d", found);
28     return 0;
29 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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SAHANA A 2024-CSE ▾

S2

Started on Tuesday, 4 November 2025, 7:16 PM

State Finished

Completed on Tuesday, 4 November 2025, 7:16 PM

Time taken 16 secs

Marks 1.00/1.00

Grade 4.00 out of 4.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 int main() {
3     int n;
4     scanf("%d", &n);
5     int arr[n];
6     for (int i = 0; i < n; i++)
7         scanf("%d", &arr[i]);
8     int k;
9     scanf("%d", &k);
10    int i = 0, j = 1;
11    int found = 0;
12    while (i < n && j < n) {
13        if (i == j) {
14            j++;
15            continue;
16        }
17        int diff = arr[j] - arr[i];
18        if (diff == k) {
19            found = 1;
20            break;
21        } else if (diff < k) {
22            j++;
23        } else {
24            i++;
25        }
26    }
27    printf("%d", found);
28    return 0;
29 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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